Zhengyang Yu

Software Engineering Student

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Education

2025 – Jun 2028 **Bachelor of Science in Computer Science**, *University of California, San Diego*, San Diego, CA
Expected Graduation: June 2028; GPA: 4.0/4.0
Relevant Coursework: Systems Programming, Computer Organization, Advanced Data Structures, Design and Analysis of Algorithms

2023 – 2025 **Associate of Science in Computer Science**, *Laney College*, Oakland, CA

Projects

2026 **Quotaquarium**, *Personal Project*
- Designed and built a C++/FreeRTOS application that authenticates multiple accounts, retrieves live data over HTTPS, and presents account state through an interactive physical display.
- Built adaptive network workers, structured runtime diagnostics, bounded persistent logs, and visible health states to make refresh behavior and failures observable.
- Secured the external-service integration with validated TLS and device-code authorization; excluded credentials, tokens, and raw API responses from logs and tested network, state, and rendering failures.

2025 **HTML to ePub Converter**, *Personal Project*
- Engineered a Python pipeline that parses HTML, validates document structure, embeds metadata, and produces standards-compliant ePub files.
- Improved reliability through modular transformations, structured error handling, and repeatable validation of generated output.

2024 **Home Assistant Presence Plugin**, *Personal Project*
- Built a Python integration that models room occupancy from entry and exit sensor events and exposes real-time state for home automation.
- Designed an event-driven REST API workflow with explicit state transitions to support testing, troubleshooting, and reliable smart-lighting behavior.

Skills

Programming C++, Python, Java, C

Agentic AI Codex for codebase exploration, change planning, iterative implementation, debugging, and build/test verification; critical review of generated changes

Backend Object-oriented design, HTTP/HTTPS, REST APIs, authorization flows, concurrent and event-driven systems

Reliability Software testing, structured logging, runtime diagnostics, failure analysis, technical documentation

Tools Git, Docker, Linux, FreeRTOS, PlatformIO, VS Code

Activities

2024–2025 Computer Consultant, Oakland Public Library - Elmhurst Branch
Supported community members with computer troubleshooting and taught basic software skills.